

Given a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, we say that f is *convex* if for all $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$ we have

$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$. This inequality is called the *convexity inequality*. It is a necessary condition for a function to be convex. However, it is not sufficient. For example, the function $f(x) = x^2$ is convex, but the function $f(x) = x^3$ is not. The function $f(x) = x^2$ is also *strictly convex*, which means that the inequality is strict for all $x \neq y$.

1.1. Convexity and optimization

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One of the main reasons why convexity is important in optimization is that it guarantees that any local minimum is also a global minimum. This is not true for non-convex functions. For example, the function $f(x) = x^2$ has a unique global minimum at $x=0$, while the function $f(x) = x^3$ has a local minimum at $x=0$ but no global minimum.

Another important property of convex functions is that they are *submodular*. This means that for any two sets A and B , we have $f(A \cup B) + f(A \cap B) \leq f(A) + f(B)$. This property is useful in many applications, such as in the design of algorithms for finding the minimum of a function.

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One of the most important applications of convexity is in the design of algorithms for finding the minimum of a function. For example, the *gradient method* is a simple algorithm that can be used to find the minimum of a convex function.

$$x_{k+1} = x_k - \eta \nabla f(x_k)$$

where x_k is the current point, η is the step size, and $\nabla f(x_k)$ is the gradient of the function at x_k . The gradient method is a simple algorithm that can be used to find the minimum of a convex function. It is based on the fact that the gradient of a convex function at a point x is a vector that points in the direction of the steepest increase of the function at x . Therefore, if we move in the opposite direction of the gradient, we will be moving in the direction of the steepest decrease of the function, which is the direction of the minimum.

Another important application of convexity is in the design of algorithms for finding the maximum of a function. For example, the *subgradient method* is a simple algorithm that can be used to find the maximum of a convex function.

$$x_{k+1} = x_k + \eta g_k$$

where x_k is the current point, η is the step size, and g_k is the subgradient of the function at x_k .

One of the most important applications of convexity is in the design of algorithms for finding the minimum of a function. For example, the *gradient method* is a simple algorithm that can be used to find the minimum of a convex function. It is based on the fact that the gradient of a convex function at a point x is a vector that points in the direction of the steepest increase of the function at x . Therefore, if we move in the opposite direction of the gradient, we will be moving in the direction of the steepest decrease of the function, which is the direction of the minimum.

Es zeigt sich, dass die Aussage $P(n)$ für $n=1$ wahr ist. Wir zeigen nun, dass die Aussage $P(n)$ für $n=k$ wahr ist, wenn $P(n)$ für $n=k-1$ wahr ist.

$$P(k) \Rightarrow P(k+1)$$

Induktionsanfang: Für $n=1$ gilt die Aussage $P(1)$. Es gilt $1 \leq 1$, also ist die Aussage $P(1)$ wahr. $\square$

Induktions Schritt: Wir zeigen nun, dass die Aussage $P(k)$ für $k \geq 1$ wahr ist, wenn $P(k-1)$ wahr ist. Es gilt $k \leq k$, also ist die Aussage $P(k)$ wahr. $\square$

Die Aussage $P(n)$ ist für alle natürlichen Zahlen n wahr. $\square$

3.3.3. Induktion über die Anzahl der Elemente einer Menge

Die Induktion über die Anzahl der Elemente einer Menge ist eine wichtige Methode, um Aussagen über die Struktur einer Menge zu beweisen. Wir zeigen nun, dass die Aussage $P(n)$ für $n=0$ wahr ist, wenn $P(n)$ für $n=0$ wahr ist.

$$P(0) \Rightarrow P(1)$$

Induktionsanfang: Für $n=0$ gilt die Aussage $P(0)$. Es gilt $0 \leq 0$, also ist die Aussage $P(0)$ wahr. $\square$

Induktions Schritt: Wir zeigen nun, dass die Aussage $P(k)$ für $k \geq 0$ wahr ist, wenn $P(k-1)$ wahr ist. Es gilt $k \leq k$, also ist die Aussage $P(k)$ wahr. $\square$

Die Aussage $P(n)$ ist für alle natürlichen Zahlen n wahr. $\square$

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